Program : **Diploma in Instrumentation Engineering**	
Course Code : **6082A**	Course Title: **Industrial IoT**
Semester : **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week : **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

- To explain the fundamentals of the emerging technology of IoT.
- To compare traditional technology and IoT.
- To describe the design methodology used in IoT.
- To explain the applications of IoT.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sensors		Sensors and transducers	3
Micro processor		Microcontroller	4

Course Outcomes:

On completion of the course, the student will be able to

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the fundamental concepts of Internet of Things (IoT)	15	Understanding
CO2	Compare IoT and M2M	13	Understanding
CO3	Explain the IoT Platform Design Methodology	15	Understanding
CO4	Explain Applications building with IoT	15	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						
CO2	2						
CO3	2						
CO4	2	3	3	3		3	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the fundamental concepts of Internet of Things (IoT)**		
M1.01	Explain the Definition and characteristics of IoT,	2	Understanding
M1.02	Describe the Physical design of IoT	4	Understanding
M1.03	Describe the Things in IoT	4	Understanding
M1.04	Explain the IoT enabling Technologies	3	Understanding
M1.05	Describe the IoT levels and deployment templates	2	Understanding
Contents: Definition and characteristics of IoT, Physical design of IoT, Things in IoT - IoT Protocols, Logical Design of IoT, IoT functional blocks, IoT communication Models, IoT communication API's, IoT enabling Technologies - Wireless sensor networks, Cloud Computing, Big Data Analytics, Communication protocols, embedded systems. IoT Levels and Deployment templates – IoT Level-1, IoT Level-2, IoT Level-3, IoT Level-4, IoT Level-5, IoT Level-6.			
CO2	**Compare IoT and M2M**		
M2.01	Explain the Machine to machine	3	Understanding
M2.02	Describe the Difference between IoT and M2M	2	Understanding
M2.03	Describe the Software defined networking	4	Understanding
M2.04	Explain the Network function virtualization	4	Understanding
	Series Test – I	1	

Contents: Introduction - M2M, Difference between IoT and M2M, SDN and NFV for IoT- Software defined networking – advantages, different types and network function virtualization – advantages.			
CO3	**Explain the IoT Platform Design Methodology**		
M3.01	Explain the IoT Design and Methodology	5	Understanding
M3.02	Describe the Specifications	2	Understanding
M3.03	Explain the Device and component integration	5	Understanding
M3.04	Explain the Development Application	3	Understanding
Contents: Introduction, IoT Design and Methodology- Purpose and requirements specification, Process specification, Domain model specification, Information model specification, service specification, IoT level specification, functional view specification, Operational view specification, Device and component integration, application development.			
CO4	**Explain Applications building with IoT**		
M4.01	Interface Arduino /Node MCU (LED, LCD different type of sensors)	5	Understanding
M4.02	Interface Raspberry PI with sensors (LED, LCD different type of sensors)	5	Understanding
M4.03	Describe the Applications building with IoT	5	Understanding
	Series Test – II	1	
Contents: Interfacing Arduino /Node MCU (LED, LCD different type of sensors)– Features of Raspberry PI Interfacing Raspberry PI with sensors (LED, LCD different type of sensors) - Applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system.			

Text / Reference:

T/R	Book Title/Author
R1	Internet of things: Shriram K Vasudevan, Abhishek S Nagarajan, RMD Sundaram
R2	Internet of Things: A Hands-on Approach”, by Arshdeep Bahga and Vijay Madisetti
R3	The Internet of Things: Enabling Technologies, Platforms, and Use Cases”, by Pethuru Raj
R4	Internet of Things by Dr. Jeeva Jose, Khanna Publishing House (Edition 2017)
R5	Internet of Things: Architecture and Design Principles, Raj Kamal, McGraw Hill

Online Resources:

Sl. No	Website Link
1	https://nptel.ac.in/noc/individual_course. php?id=noc17-cs22
2	https://www.raspberrypi.org/blog/getting-started-with-iot/
3	https://www.arduino.cc/en/IoT/HomePage
4	https://www.microchip.com/design-centers/internet-of-things
5	https://learn.adafruit.com/category/internet-of-things-iot
6	http://esp32.net/